

Plotting, Ordering and Comparing Real Numbers

MA.8.NSO.1.2

I can compare, order and plot every kind of real number in this unit.

Nothing new today.

This is the rehearsal. The assessment is next.

Warm-Up

On your own. Four minutes.

Same and different

What is the same about these two, and what is different?

$$\sqrt[3]{216} \qquad \sqrt{36}$$

Review

I. Is it rational? Check the radicand, not the radical

- A. **Rational:** any ratio of two integers — every fraction, every whole number, every decimal that stops or repeats.
- B. $\sqrt{}$ of a perfect square and $\sqrt[3]{}$ of a perfect cube are rational. They are integers.
- C. Anything else under a $\sqrt{}$ or a $\sqrt[3]{}$ is irrational. So is π , and π times any NON-ZERO fraction.

Review

II. Convert first. Then order by position

- A. **Convert every number to a decimal, and write the decimals down.**
- B. Exact where you can. Nearest tenth where you cannot.
- C. **Then order by position: further left is less, further right is greater. Always.**
- D. **Watch the negatives. -7 is LESS than -6.28 .**

Review

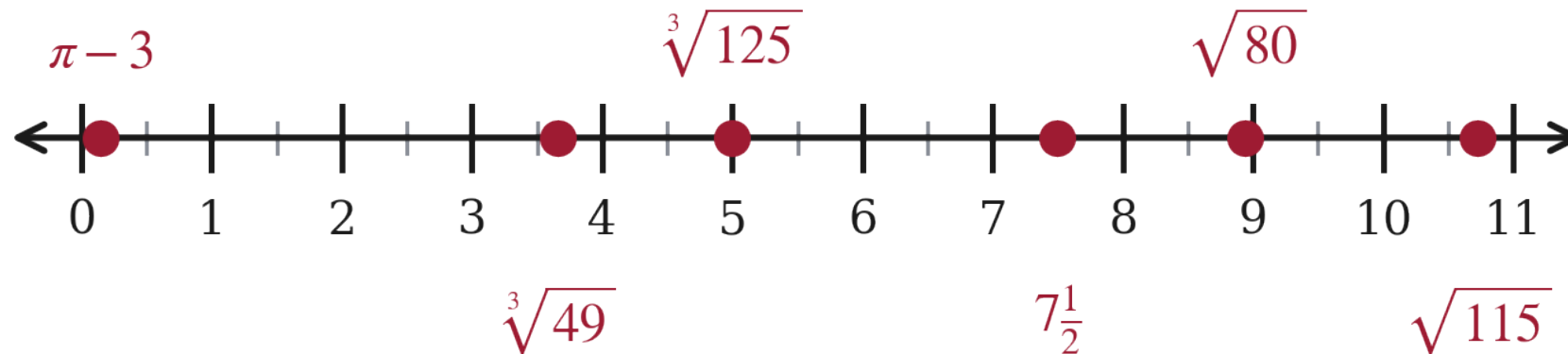
III. Plot: integers first, then the tenth

- A. **Check the tick points before you start. They are not always worth 1.**
- B. Find the two integers it sits between, put a light mark between them, then slide the point to the tenth.
- C. **Label every point. An unlabeled dot earns nothing.**
- D. If two points are close together, put one label above the line and one below.

Review

Six points on one line — question 11 of the circuit.

III. Plot: integers first, then the tenth



Example 1

Four comparisons. One of them is an equality.

Left		Right	Decimals
-5π		$-\sqrt{225}$	
$\sqrt[3]{125}$		$\sqrt{100} \div 2$	
-7		$-\sqrt[3]{49}$	
$29/3$		$\sqrt{80}$	

Example 1

The decimals.

Left		Right	Decimals
-5π		$-\sqrt{225}$	-15.71 and -15
$\sqrt[3]{125}$		$\sqrt{100} \div 2$	5 and 5
-7		$-\sqrt[3]{49}$	-7 and -3.66
$29/3$		$\sqrt{80}$	9.67 and 8.94

Example 1

The symbols.

Left		Right	Decimals
-5π	$<$	$-\sqrt{225}$	-15.71 and -15
$\sqrt[3]{125}$	$=$	$\sqrt{100} \div 2$	5 and 5
-7	$<$	$-\sqrt[3]{49}$	-7 and -3.66
$29/3$	$>$	$\sqrt{80}$	9.67 and 8.94

Example 2

Rational or irrational? Work each radical out first.

10.2	$11\frac{1}{3}$	$\sqrt[3]{64} + 8$	$-\pi + 16$	$\sqrt{190}$

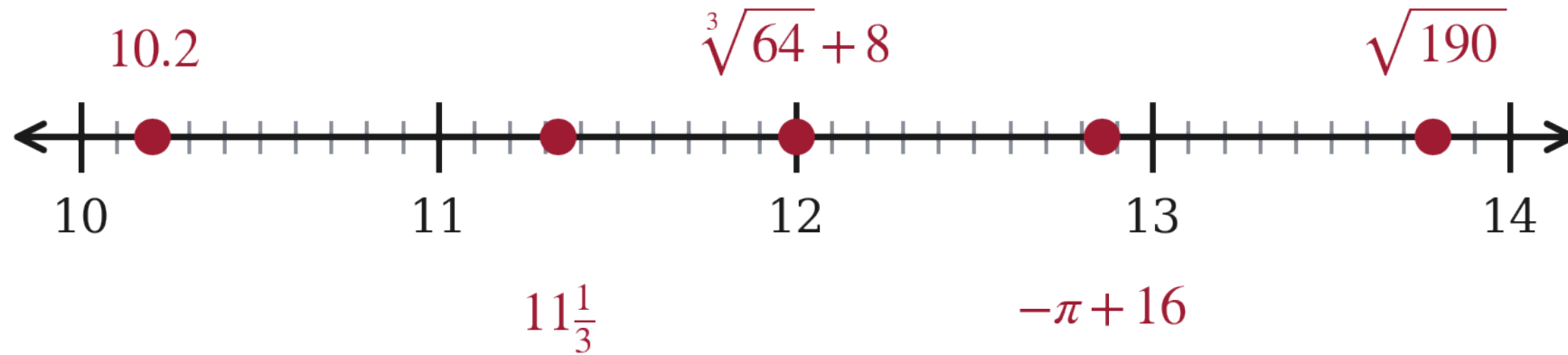
Example 2

Sorted, and their decimals.

10.2	$11\frac{1}{3}$	$\sqrt[3]{64} + 8$	$-\pi + 16$	$\sqrt{190}$
rational	rational	rational	irrational	irrational
10.2	11.33	12	12.86	13.78

Example 2

Plotted.



Example 3

Sara's reasoning. Read it carefully.

“11 is between 9 and 16, which are both perfect squares.

So $\sqrt[3]{11}$ lies between 3 and 4, and it is closer to 3.”

What is RIGHT about what she did?

Example 3

And what is wrong.

What Sara did	What she should have done
$9 < 11 < 16$ perfect SQUARES	$8 < 11 < 27$ perfect CUBES
between 3 and 4	between 2 and 3

Example 3

What it costs.

Sara's answer	The right answer
$\sqrt[3]{11} \approx 3.3$	$\sqrt[3]{11} \approx 2.2$

Whiteboards • Question 1 of 9

Boards up on three.

$$\sqrt{36} \qquad 6$$

Compare the two numbers.

Write <, > or =.

Whiteboards • Question 1 of 9

Answer.

$$\sqrt{36} \quad 6$$

6.00 and 6.00

Answer: =

Whiteboards • Question 2 of 9

Boards up on three.

$$\sqrt{37} \qquad 6$$

Compare the two numbers.

Write <, > or =.

Whiteboards · Question 2 of 9

Answer.

$$\sqrt{37} \qquad 6$$

6.08 and 6.00

Answer: >

Whiteboards • Question 3 of 9

Boards up on three.

$$-\sqrt{37} \qquad -6$$

Compare the two numbers.

Write <, > or =.

Whiteboards · Question 3 of 9

Answer.

$$-\sqrt{37} \qquad -6$$

−6.08 and −6.00

Answer: <

Whiteboards • Question 4 of 9

Boards up on three.

$$\sqrt[3]{37} \qquad 6$$

Compare the two numbers.

Write <, > or =.

Whiteboards • Question 4 of 9

Answer.

$$\sqrt[3]{37} \qquad 6$$

3.33 and 6.00

Answer: <

Whiteboards • Question 5 of 9

Boards up on three.

$$\sqrt[3]{216} \qquad 6$$

Compare the two numbers.

Write <, > or =.

Whiteboards • Question 5 of 9

Answer.

$$\sqrt[3]{216} \quad 6$$

6.00 and 6.00

Answer: =

Whiteboards • Question 6 of 9

Boards up on three.

$$\sqrt[3]{216}$$

$$\sqrt{36}$$

Compare the two numbers.

Write <, > or =.

Whiteboards • Question 6 of 9

Answer.

$$\sqrt[3]{216}$$

$$\sqrt{36}$$

6.00 and 6.00

Answer: =

Whiteboards • Question 7 of 9

Boards up on three.

$$\frac{22}{7}$$

$$\pi$$

Compare the two numbers.

Write <, > or =.

Whiteboards • Question 7 of 9

Answer.

$$\frac{22}{7}$$

$$\pi$$

3.142857... and 3.141592...

Answer: >

Whiteboards • Question 8 of 9

Boards up on three.

$$\sqrt[3]{216}$$

$$\pi$$

Compare the two numbers.

Write <, > or =.

Whiteboards • Question 8 of 9

Answer.

$$\sqrt[3]{216}$$

$$\pi$$

6.00 and 3.14

Answer: >

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Put these in order, least to greatest:

6	$-\sqrt{37}$	$\sqrt[3]{37}$
$\sqrt{37}$		

Write your answer in sentences.

This one is written work. Say how you decided.

Whiteboards • Question 9 of 9

Answer.

Put these in order, least to greatest: $-\sqrt{37}$ $\sqrt[3]{37}$
 6 $\sqrt{37}$

$$-6.08 < 3.33 < 6.00 < 6.08$$

Answer: $-\sqrt{37}$, $\sqrt[3]{37}$, 6 , $\sqrt{37}$

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.